



What do four sensors miss?

Spatial observability of low-cost plantar sensor arrays

Full research title

Establishing the observability and reliability limits of low-cost plantar sensor arrays for wearable gait characterisation

CATEGORY

Engineering & Technology

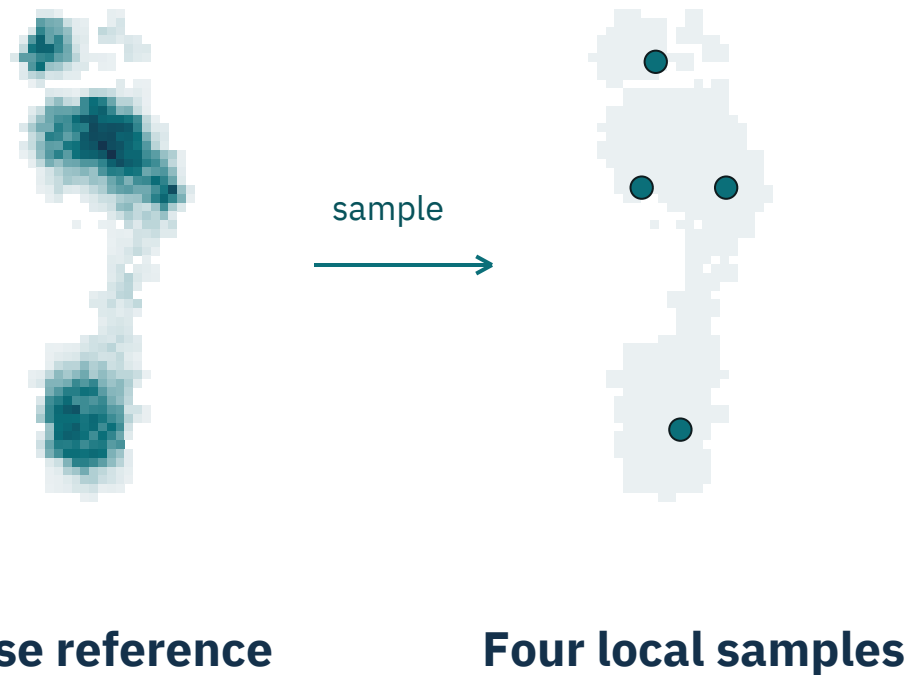
LEVEL

UG

CODE NO.

PROBLEM STATEMENT

Missing pressure limits gait characterisation



A load shift can occur between sensors

Pressure outside the discs has no direct measurement channel.

Engineering impact

Whole-foot load distribution and centre-of-pressure estimates then depend on assumptions about the missing field.

Same BF right-foot example: participant 002, W1, step 1. Discs use the assumed 12.7 mm geometry.

RESEARCH GAP + OUR CONTRIBUTION

Characterisation is the research contribution

GAP FOR THIS INSTRUMENT

Spatial coverage

Unknown coverage of the assumed sparse layouts.



OUR CONTRIBUTION

R2 / completed

Reproducible overlap analysis on a fixed public-data subset.

Endurance

Sensitivity change under repeated load is unmeasured.



R1 / planned

A sensitivity-versus-cycles study with repeated calibration.

Measurement validity

Loss of validity has no validated detection rule.



R3 / proposed

Test a reporting gate with deliberately aged sensors.

The insole is the instrument. The completed contribution is the R2 subset analysis.

AIM & OBJECTIVES

Define the instrument’s measurement limits

R2

COMPLETED SUBSET

Spatial coverage

Quantify area and load overlap across five fixed layout models.

Output: participant means and ranges

R1

PLANNED

Endurance

Track sensitivity and zero offset against repeated load cycles.

Output: degradation curve

to be measured

R3

PLANNED

Measurement validity

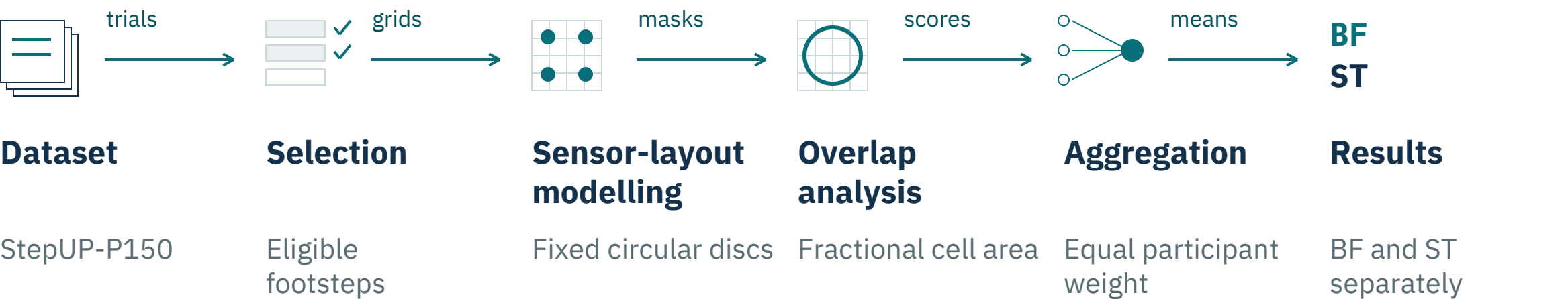
Test drift detection using sensors aged in R1.

Output: detection rate, false alarms, latency

to be measured

METHODOLOGY

R2 analysis workflow



Aggregation order



SYSTEM / MODEL STATUS

Implementation and validation status

IMPLEMENTED/ANALYZED

R2 computational pipeline

Selection, layout modelling and overlap analysis run on archived pressure data.

Existing acquisition firmware

ESP32 with four FSR402 channels, MPU6050 and heel piezo.

PENDING

Integrate niva_signal.h fixes, then verify acquisition and calibration.

PROPOSED / VALIDITY GATE

Drift checks

Baseline drift
Re-zero divergence
Channel consistency

check outcome

VALID
Report

DEGRADED
Suppress

INVALID
Recalibrate

PENDING: gate integration and R3 tests.
Detection performance: to be measured.

EXPERIMENTAL SETUP

How the 576 footsteps were selected

12

×

2

×

2

×

12

=

576

Participants

Conditions

Feet

Steps

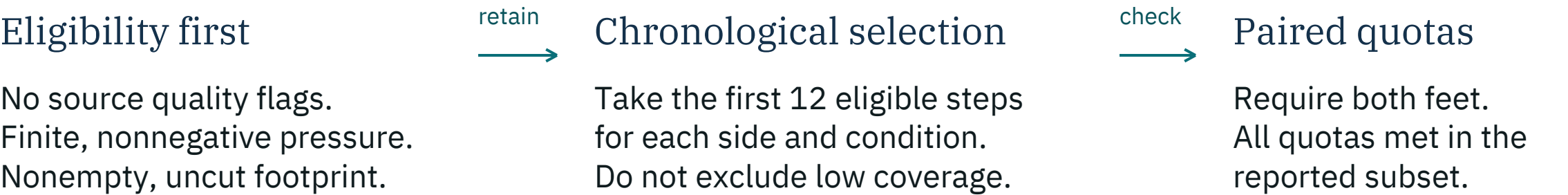
Selected steps

IDs 002-013

BF + ST
Preferred speed (W1)

Left + Right

Per foot / condition



MAIN RESULTS / FOUR-SENSOR MODEL

Most loaded area remains unobserved

3 . 55%

Barefoot (BF)

2 . 18%

Sneaker (ST)

Mean loaded area inside the four model discs

Four-sensor model	Barefoot	Sneaker
Participant range, area (%)	2 . 76–4 . 25	1 . 66–2 . 81
Mean observed load (%)	3 . 90	2 . 09

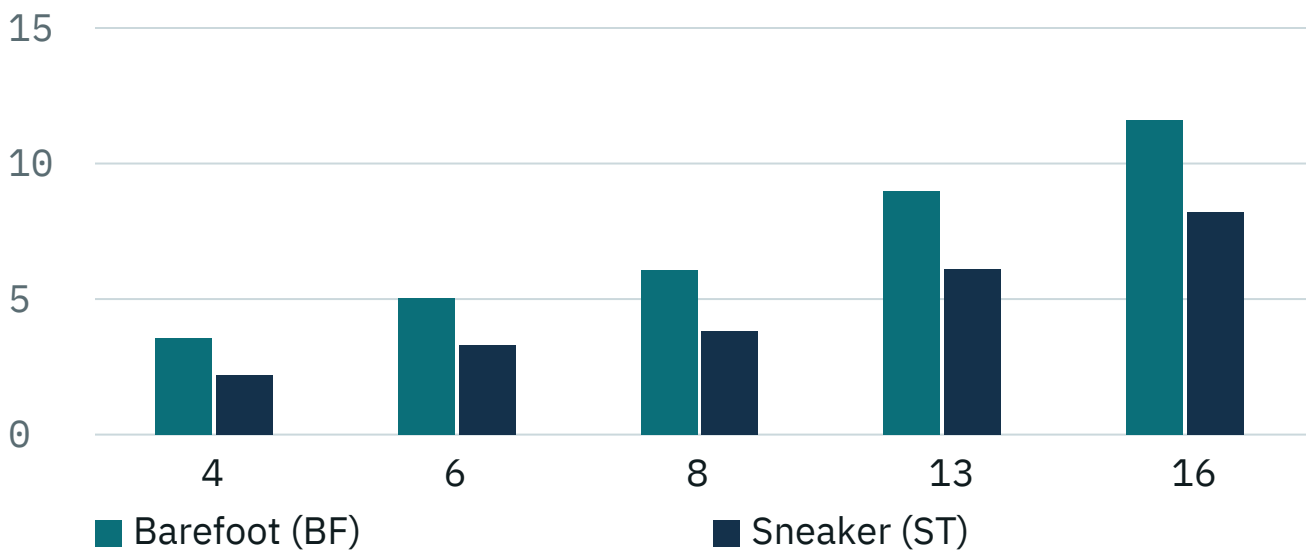
Sparse samples leave most of the reference field without direct measurement.

Means of 12 participant summaries. Ranges are participant minima–maxima, not confidence intervals.

RESULTS & COMPARISON

Layout comparison with participant variability

Mean loaded area observed (%)



Sensor count in each layout model

Participant range, area (%)

Count	BF	ST
4	2.76–4.25	1.66–2.81
6	4.25–5.69	2.69–4.02
8	4.91–6.51	3.08–4.53
13	6.95–10.43	5.20–7.14
16	10.12–13.66	6.96–9.62

12 participant means per condition.
Ranges are not confidence intervals.

Count and position both change. The chart compares layouts, not a sensor-count effect.

APPLICATIONS & LIMITATIONS

What the study establishes

ESTABLISHES

A reproducible coverage benchmark

Quantifies spatial blind area for fixed layouts on a declared pressure-data subset.

A basis for measurement decisions

Shows where whole-foot interpretations need additional validation.

BF and ST are floor-interface data. No clinical conclusions follow from this study.

DOES NOT ESTABLISH**In-shoe performance**

Next: measured geometry and an in-shoe reference.

Sensor endurance or gate accuracy

Next: R1 cyclic loading, followed by R3 drift tests.

A minimum sensor count

Next: a controlled placement study with declared error targets.

CONCLUSION & FUTURE SCOPE

Spatial coverage is limited under this protocol

The four-sensor model leaves most of the loaded reference surface outside its active footprints.

R1 / endurance

Cyclic loading and repeated calibration
Sensitivity change: to be measured

R3 / validity

Use aged sensors from R1 to test the gate
Detection performance: to be measured

Next prerequisite: integrate and verify the acquisition fixes before hardware measurement.

REFERENCES

Sources and reproducibility

- 01 **Larracy et al. StepUP-P150 dataset paper**
Scientific Data 12, 1415 (2025). doi:10.1038/s41597-025-05792-1
- 02 **StepUP-P150 research dataset**
Federated Research Data Repository. doi:10.20383/103.01285
- 03 **Niva predeclared R2 sampling plan**
docs/R2_sampling_plan.md · commit 799f322
- 04 **R2 analysis and outcome archive**
tools/r2_study.py · docs/r2/run_20260912T161228Z/
- 05 **Niva research framework and work package**
Project documents: methods, claims, hazards and planned experiments



Thank you

Questions & discussion

Observability limits. Endurance testing. Validity reporting.